FACULTY OF COMPUTING

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SECP3213 – BUSINESS INTELLIGENCE

SECTION 1

PHASE 1: INTRODUCTION (PROPOSAL STAGE)

PROJECT TITLE:

Integrated Retail Sales and Customer Analytics for Business Decision Support Using Alteryx and Power BI

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1. PROBLEM BACKGROUND

1.1 BUSINESS CONTEXT

The rapid development of e-commerce and the retail industry have generated massive amounts of transaction and customer-related data. Enterprise needs to effectively manage sales performance, customer purchasing behaviour, product demand and profitability in order to maintain competitiveness in the market. However, many organizations face difficulties in turning raw business data into meaningful insights for decision-making.

Business intelligence (BI) tools such as Alteryx and Power BI enable organizations to perform data integration, cleaning, conversion and visualization to support strategic business decision-making. By analysing retail sales and customer analysis, enterprise can more effectively identifying purchasing trends, customer segmentation, profitable products and regional sales performance.

The project focuses on integrating and analysing multiple retail and customer-related data sets to develop business intelligence solutions that support business decision-making through interactive dashboards and data-driven insights.

1.2 IMPORTANCE OF THE PROBLEM

Due to the increasing volume and complexity of business data, retail and e-commerce enterprise often face difficulties in understanding customer purchasing behaviour and overall sales performance. Without proper analysis, it may be difficult for organizations to identify profitable products, understand customer preference and improve customer satisfaction. Poor data management can also lead to ineffective business strategies, inaccurate forecasts and missed business opportunities.

Therefore, business intelligence (BI) solutions are very important to convert raw data into meaningful decision-making insights. By using Alteryx, organization can effectively perform data preparation processes like data cleaning, processing missing values and data conversion. Power BI helps visualise processed data through interactive dashboards and reports, enabling enterprise to monitor key performance indicators, identify trend and support strategic decision-making.

This project is important because it shows how multiple retail-related data sets can be used together to support retail sales and customer analysis. The dataset will support analysis areas such as sales performance, customer segmentation, profitability analysis and purchasing behaviour to provide valuable business insights.

1.3 EXPECTED IMPACT

The expected impact of this project is to understand retail business performance more clearly through data-driven analysis. Proposed BI solutions can help businesses more effectively identify sales trends, profitable products, customer segmentation and reginal performance.

In addition, the project can support better decision-making through interactive dashboard and key performance indicator monitoring. By analysing customer behaviour and sales performance, enterprise can improve their marketing strategy, customer positioning and operational efficiency.

Overall, the project is expected to provide meaningful business insights and support strategic business decision-making through data visualisation and business intelligence.

2. OBJECTIVES

The objectives of this project are:

1. To integrate multiple retail-related datasets using Alteryx Designer for business intelligence analysis.
2. To perform data cleaning and transformation processes, including handling missing values, removing duplicate records, standardizing data formats。
3. To analyse retail sales performance, customer behaviour, product profitability to identify meaningful business insights.

4. To develop interactive dashboards using Microsoft Power BI for data visualization and business decision support.

3. DATASETS DESCRIPTION

3.1 DATASET 1 - Brazilian E-Commerce Public Dataset by Olist

This dataset contains e-commerce order information from Olist, a Brazilian online marketplace. It includes multiple CSV files related to customers, orders, order items, payments, products, sellers, reviews, and product category translation. This dataset is structured data because it is provided in CSV format with rows and columns.

The key variables include:

Variable	Description	Variable	Description
order_id	Unique order identifier	payment_value	Total payment amount
customer_id	Unique customer identifier for each order	freight_value	Shipping cost
order_purchase_timestamp	Purchase date and time	review_score	Customer review rating
product_id	Unique product identifier	customer_city	Customer location
seller_id	Unique seller identifier		

3.2 DATASET 2 - Global Superstore Dataset

This dataset contains retail transaction records from a global superstore business. It includes sales, profit, order date, shipping date, customer segment, product category, sub-category, region, country, shipping mode, and quantity sold. This dataset is also structured data because it is provided in CSV format.

The key variables include:

Variable	Description	Variable	Description

Order.Date	Date of transaction	Quantity	Quantity sold
Customer.ID	Unique customer identifier	Region	Sales region
Category	Product category	Segment	Customer segment
Sales	Total sales amount	Ship.Mode	Shipping method
Profit	Profit earned		

3.3 DATASET 3 - Customer Personality Analysis

This dataset contains customer demographic information, income level, spending behaviour, purchase frequency, and marketing campaign responses. The dataset is structured data and is provided in CSV format.

The key variables include:

Variable	Description	Variable	Description
ID	Customer's unique identifier	Recency	Number of days since customer's last purchase
Year_Birth	Customer's birth year	MntMeatProducts	Amount spent on meat
Education	Customer's education level	MntWebPurchases	Number of purchases made through the company’s website
Marital_Status	Customer's marital status	Response	Campaign response
Income	Customer's yearly household income		

4. PROPOSED BI APPROACH

4.1 TOOLS USED

The main tools used in this project are **Alteryx Designer** and **Microsoft Power BI.**

Alteryx Designer will be used for the data preparation and workflow development. It allows project team to import multiple datasets, clean raw data, transform fields, join related tables, create calculated variables, and export cleaned data for visualization use. Alteryx is mostly suitable for this project because the selected datasets contain multiple files and require data preparation before doing any analysis and development use.

Microsoft Power BI will be used for data visualization, summarized report and dashboard development. It allows project team to create interactive dashboards, KPI cards, charts, slicers, filters, business reports and so on. Power BI is mostly suitable for this project because it can present business insights in a visual format that is easy for project team to understand and act as presentation use.

The combination of Alteryx and Power BI supports the full BI process from data preparation to insight generation. For the examples, Alteryx will handle the backend data processing while Power BI will focus on dashboard design and business storytelling such a win-win situation.

4.2 DATA INTEGRATION, CLEANING, AND TRANSFORMATION METHODS

The data preparation process will initiate by importing the selected datasets into **Alteryx Designer**. Due to the datasets come from different public sources, direct joining will only be applied when common and meaningful fields exist within the same dataset. For the example, the Olist dataset contains multiple related CSV files that can be joined using keys such as order_id, customer_id, and product_id.

Besides, data integration will be performed at an analytical level instead of treating them as one single database for the datasets from different sources. The data analysis will compare common business dimensions such as sales, profit, customer segment, product category, region, purchase behaviour, and marketing response. This approach allows the project to build a meaningful retail BI solution without if all datasets belong to the same organization.

After that, the data cleaning process will also include removing duplicate records, handling missing values, correcting inconsistent data formats, standardizing column names, and removing unnecessary fields. For a instances, the missing income values in the marketing dataset may need to be handled before doing the customer segmentation analysis. The date fields such as order date and shipping date will also be converted into a consistent format for time-series analysis.

After doing the cleaning process, the data transformation process will be proceeded include filtering relevant records, aggregating sales and profit values, create calculated fields, and categorize customers or products. Examples of calculated fields such as total sales, profit margin, customer age, total spending, purchase frequency, and campaign response rate. These transformed fields will be useful for dashboard creation and the business analysis.

After all data preparation process is completed successfully, the cleaned datasets will be exported from Alteryx and imported into **Power BI** for visualization. Now, **Power BI** will then be used to create dashboards that show business performance, customer insights, and marketing effectiveness.

4.3 PLANNED ANALYSIS

No.	Planned Analysis	Description	Expected Business Insight
1	Sales Trend Analysis	It used to analyse monthly and yearly sales performance using order date and sales data.	Identify sales growth patterns, seasonal trends, and periods of high and low sales.
2	Product Performance Analysis	It used to compare product categories and sub-categories based on the sales, quantity, reviews, and the profit.	Identify top products and low products that may need improvement.
3	Profitability Analysis	It used to analyse the sales, profit, profit margin, and shipping cost across product categories and customer segments.	Identify profitable and low-profit areas to support pricing, cost control, and product strategy.

4	Regional Performance Analysis	It used to compare sales and profit across customer locations, regions, countries, and states.	Identify strong markets and weaker regions that require more business attention.
5	Customer Segmentation Analysis	Grouping the customers based on income, age, education, spending pattern, purchase frequency, and campaign response.	Identify high-value customer groups and support more targeted customer strategies.
6	Marketing Campaign Analysis	It used to evaluate customer responses to marketing campaigns using campaign acceptance and response fields.	Understand which customer groups are more likely to respond to limited offer and its promotions.

4.4 PROPOSED VISUALIZATIONS

The project will use several visual forms in Power BI to present the analysis clearly and effectively.

Visual Form	Purpose
Line chart	It shows monthly or yearly sales trends.
Bar chart	It compares sales, profit, and quantity across categories.
Map visualization	It displays regional or country-based performance.
Donut chart	It shows customer segment or category distribution.
Scatter plot	It analyses relationships such as sales vs profit.
Treemap	It shows product category contribution.
Stacked column chart	It compares campaign responses across groups.

5.0 REFERENCES

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